

CAPSTONE PROJECT REPORT

Job Market Data Pipeline

A trusted, repeatable view of hiring demand in Saudi Arabia

6

job sources, all through APIs

52,937

raw records collected

19,148

unique jobs after cleaning

288

automated data tests

Six job sources → Azure Data Lake → Snowflake → dbt star schema → Power BI

Team A (Group A)

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Data collected 9–28 September 2026 · final pipeline run 20260929T115551Z

SUMMARY

The project at a glance

Job postings in Saudi Arabia are spread across employer career pages and job-search sites, each with its own format, spelling and gaps, and the same job often appears several times. Nobody can answer a simple question such as **"how many jobs are open in Riyadh?"** from that raw data.

We built a data pipeline that collects postings from six public sources, keeps every raw record, cleans and standardizes them, recognizes the same job across sources, and delivers one tested dataset that a dashboard can read directly. One command reruns the pipeline for the four employer boards, so the dataset can be refreshed every week; the search-engine campaigns run separately because of their API quotas.

20,490

unique listings from 6 sources

19,148

unique jobs after merging

3,004

employers identified

1,008

jobs found on 2+ sources

What the data says (September 2026)

Half of all jobs are in the Riyadh region (50.4%, 9,649 jobs); the Eastern Province is second with 4,114.

Engineering and Construction (4,493) and **Business and Finance** (4,221) are the largest role families.

Qiddiya, AccorHotel and AtkinsRéalis are the largest direct employers, led by giga-project, engineering and hospitality companies.

93.1% of jobs are full-time, where the posting states it.

English (25.9%) and Arabic (16.4%) are the most requested skills, ahead of procurement and project management.

How this report answers the brief

Question	Section
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What we built, and who it is for

What we built

- **A collection layer** that pulls postings from four employer job boards (Ashby, Greenhouse, SmartRecruiters, Workable) and two job-search engines (Jooble, JSearch). All six are APIs; no website scraping, and sites whose terms forbid automated collection (LinkedIn, Indeed, Bayt and others) were excluded.
- **A data lake** in Azure (ADLS Gen2) that stores every file as collected, never edited or deleted; the employer-board scripts keep only Saudi postings.
- **A warehouse model** in Snowflake, built with dbt: raw → cleaned → matched → a star schema of one fact table, eight dimensions and a skills bridge.
- **A quality net** of 288 automated tests and reconciliation checks that stop the run before a wrong number can be published.
- **Outputs** for people and tools: a Power BI dashboard, Parquet files in the data lake, and CSV files of the ten final tables.

Who it is for

Audience	What they get
Job seekers and university career centres	Where the jobs are, which employers hire most, and which skills to build
HR and workforce planners	Demand by region, role family and experience level; which roles open and close each week
Policy and labour-market analysts	A repeatable, documented view of the private job market that can be tracked over time
Data analysts	A clean star schema in Snowflake: every business question is one join, ready for Power BI

Nine business questions

The model was designed backwards from the questions it must answer:

#	Question
Q1	How many unique jobs were open in Saudi Arabia during the period?
Q2	Which regions and cities have the most jobs?
Q3	Which employers hire the most (agencies excluded)?
Q4	Which role families and job categories are largest?
Q5	What share of jobs is full-time, part-time, contract...?
Q6	Which experience levels are most requested in each region?
Q7	How many new jobs appear each week?
Q8	How long were jobs listed before they disappeared?
Q9	Which skills are mentioned most?

2 · HOW IT WORKS

How it works

The pipeline follows the **ELT** pattern: load the raw data first, transform it inside the warehouse. API quotas are small (a Jooble key allows 500 requests for life), so keeping the raw data means any rule can be improved and the dataset rebuilt in minutes without calling the sources again.

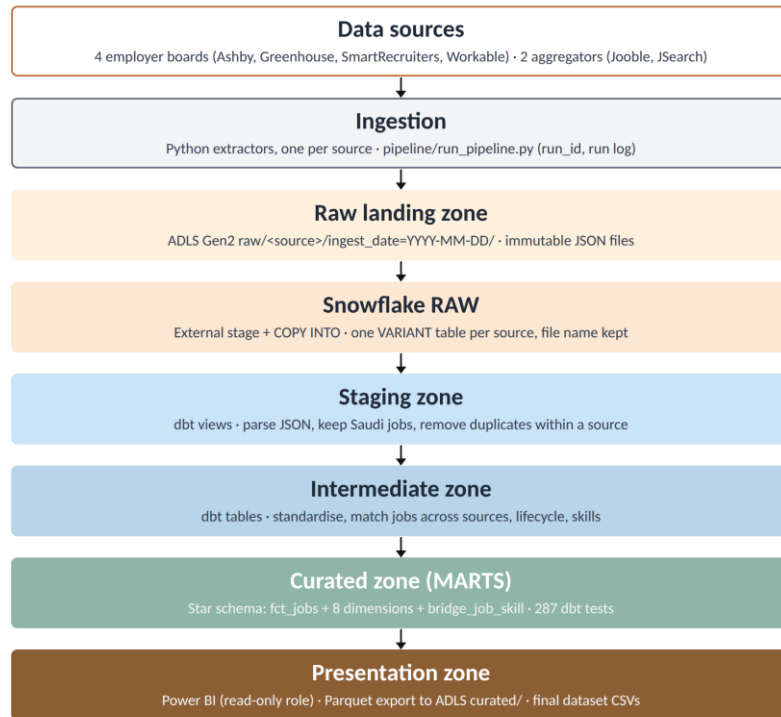


Figure 1. Data flow, from the sources to the dashboard

Step	What happens	Why it matters
1 Collect	Python scripts call each API; search engines run a fixed matrix of city, keyword and date queries; JSearch keys rotate when a quota runs out	Coverage without scraping; a query cut short is re-run by name
2 Land	Every response is saved as JSON in ADLS, one folder per source and date	An untouched audit trail: 3,007 files
3 Load	Snowflake COPY INTO loads each file once into a raw table	Re-running never duplicates data
4 Transform	dbt cleans, standardizes, matches and models the data	Rules are SQL in Git: reviewable and testable
5 Test and publish	Tests run on every build; the tables export to Parquet and feed Power BI	A failed check stops the run, so a wrong number is never published

From raw records to trusted jobs

Most of the work is turning noisy records into jobs that are counted once. Search engines return the same posting many times across overlapping queries, and one job is often advertised on several sites.

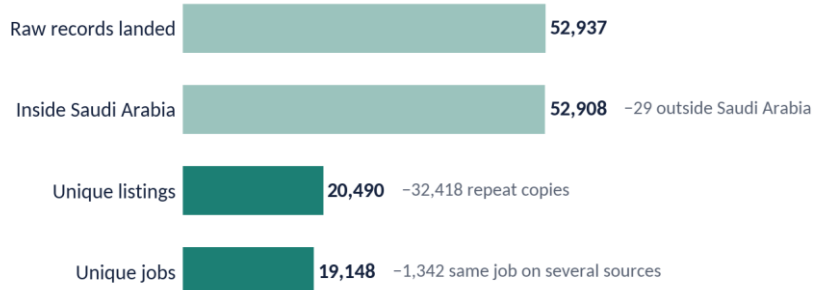


Figure 2. From 52,937 raw records to 19,148 unique jobs

Standardize

Reviewed lookup tables ("seeds") map 223 English and Arabic city spellings to one city and region (جدة → Jeddah), 742 company spellings to one employer, and 171 title keywords to 19 job categories in 7 role families.

Keep quality visible

A missing value becomes an explicit "Unknown", never a blank, so every report can show how complete its answer is.

Recognize the same job

Listings are compared only within the same employer and city. Identical normalized titles merge first; titles worded differently merge when their word overlap (Jaccard score) reaches 80. Two postings from the same publisher are never merged, because an employer that posts twice means two openings.

Example: "Sr. Project Engineer – Riyadh" on JSearch and "Senior Project Engineer" on Jooble for the same company become one job.

The final dataset is a star schema: **fact_jobs** (one row per job, with its lifecycle dates and counts) joined to eight dimensions and a skills bridge. Every business question is a single join from the fact table.

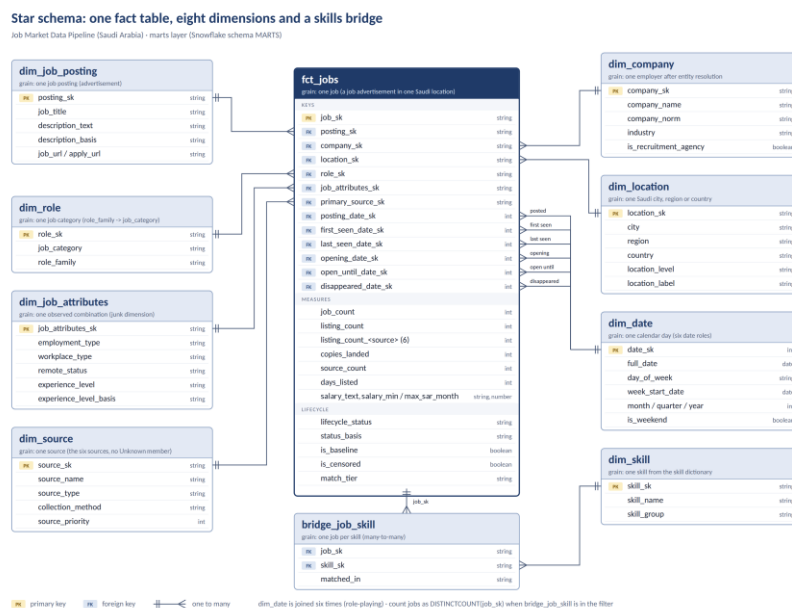


Figure 3. The star schema in the Snowflake MARTS layer

The techniques behind cleaning and matching

Every value in the dataset can be traced to a rule. These are the text-processing techniques the transformation layer uses, all written in SQL, reviewed in Git and covered by tests:

Component	Technique	Where it is used
Fuzzy text matching	Jaccard word overlap on normalized job titles (Jaro-Winkler was also scored and rejected)	Recognizing the same job advertised with different wording
Human-labelled evaluation	88 candidate pairs labelled "same job / different job"; precision and recall computed for every threshold, as in a supervised evaluation	Choosing the matching threshold (Jaccard 80: precision 0.83–0.92) and rejecting Jaro-Winkler
Text normalization	HTML stripping, case and punctuation rules, abbreviation expansion (sr → senior), removal of hiring noise	Title and company keys, cleaner descriptions
Entity resolution	Company alias dictionary (742 spellings); recruitment agencies flagged	One employer per real company
Information extraction	Dictionary of 133 skill keywords (125 skills, 12 groups) matched in descriptions; salary parsed from free text and converted to SAR per month	Skills bridge (Q9); salary fields
Rule-based classification	Title keywords → job category and role family; seniority words → experience level, with the basis recorded	Q4 and Q6

How we checked that the rules give the right answer

Tests prove that a rule holds; they cannot prove that the rule is right. So the team also compared the output with the real postings:

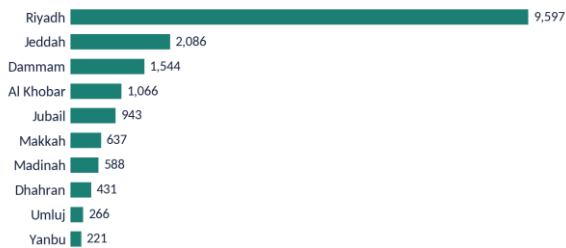
Check	Sample	Result
Fuzzy merges	All 20, read by title	18 clearly the same job
Matching threshold	88 labelled pairs	Jaccard 80: precision 0.83–0.92; Jaro-Winkler rejected
Same posting on JSearch and Jooble	All 49 match keys	Same job in 49 of 49; merged
Disappeared jobs	11 opened in a browser	9 closed on the employer site
Experience level from titles	20 jobs	18 reasonable
Salary parsing	Samples plus every extreme value	Parsing correct; one rounding issue found and fixed

3 · RESULTS

What the data shows

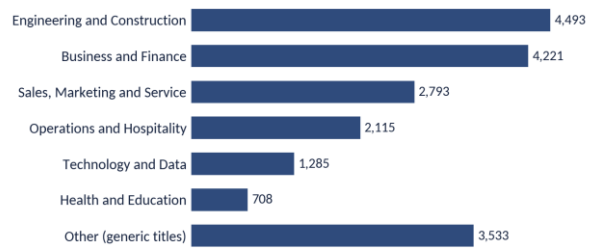
Observation period: September 2026. Counts are unique jobs after merging.

Jobs by city (top 10)



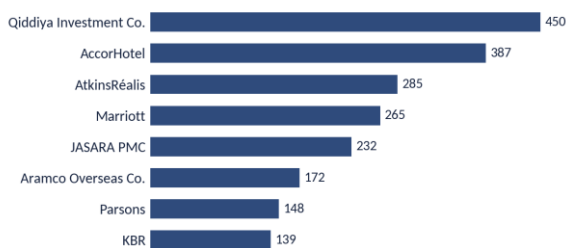
Q1-Q2 · Where the jobs are. 19,148 unique jobs. The Riyadh region holds half of them (9,649); the Eastern Province, with Dammam, Al Khobar, Jubail and Dhahran, is second (4,114); Makkah region third.

Jobs by role family



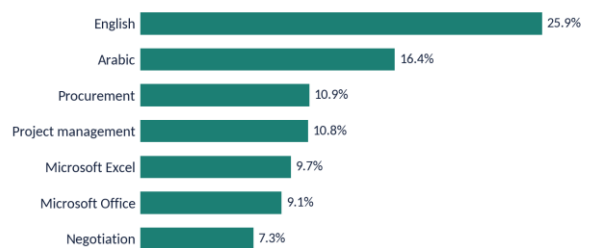
Q4 · What kind of roles. Engineering and Construction and Business and Finance lead. 18.5% of jobs have titles too generic to classify and are shown openly as "Other".

Top employers (agencies excluded)



Q3 · Who is hiring. Giga-project, engineering and hospitality employers lead. Recruitment agencies and job boards are excluded, because they advertise on behalf of others.

Most requested skills (% of jobs with a full description)



Q9 · Skills. Among 7,430 jobs with a full description, 5,554 mention at least one of 125 tracked skills; languages lead.

3 · RESULTS

Employment type, experience and weekly movement

93.1%

Q5 · full-time (6,276 of 6,743 jobs that state a type)

Mid-Senior

Q6 · top level in Riyadh and the Eastern Province

174

Q7 · new openings on employer boards

46 days

Q8 · median time listed before a job disappeared (week of 20 Sep)

Q6 · Experience level by region. Counted where the source states the level (Workable and SmartRecruiters). Mid-Senior leads in Riyadh (502 of 906 jobs) and the Eastern Province (177 of 354); Entry level leads in Makkah (105 of 259), Tabuk (105 of 185) and Madinah (37 of 57).

Q7-Q8 · Weekly movement on employer boards. Employer boards list every open job of a company, so a job that leaves the board has disappeared. This lets us count openings and closures week by week:

Week starting	New openings	Disappeared	Median days listed
20 September	150	141	46
27 September*	24	36	30.5

* Collected on 27–28 September: the first week in which all six sources were fully pulled.

Power BI

The dashboard imports the MARTS schema only and refreshes through a read-only Snowflake role. Its eight pages answer the nine questions, with slicers for region, role family, source type and employment type. Every measure was checked against the final datasets (43 of 43 checks).

Where to find the work

- Code, Snowflake worksheets, setup and run instructions: [02_code/](#) (see its README).
- The ten final tables: [02_code/01_data/final_datasets/](#) as CSV, and ADLS [curated/](#) as Parquet.
- Dashboard: [02_code/02_src/powerbi/](#), with a screenshot of each page in [02_code/03_assets/](#).
- Data model and every design decision: [02_code/02_src/dbt/data_modeling/data_model.md](#).

4 · WHAT WORKED

What worked well

- **APIs only, no scraping.** Six sources collected legally and reliably; 17 candidates were evaluated before choosing.
- **Raw data kept untouched.** Every rule change was rebuilt from the data lake in minutes, with no extra API calls against tight quotas.
- **One command, safe to repeat.** The runner executes extract → upload → load → test → build → export, logs a run ID, and stops at the first failure. Loading the same file twice has no effect.
- **Tests that block bad data.** 288 data tests run on every build (323 dbt checks in all, with seeds and models) and passed with no warnings and no errors; all 19 reconciliation checks passed. Row counts reconcile from raw data to the final fact table.
- **Hand checks against real postings.** All 20 fuzzy merges were read (18 clearly the same job); 11 disappeared jobs were opened in a browser; salary parsing and title rules were sampled.
- **A clear, simple model.** One fact table answers all nine questions, with Unknown members instead of blanks, so completeness is always visible.

5 · LIMITS

What did not work, or is still incomplete

Some approaches were tested and rejected, and some parts are limited by what the sources publish:

Area	What happened	How we handled it
A "not seen for 7 days = closed" rule	On search-engine results it would have marked 89.3% of listings as closed, against 5.2% on employer boards, and every test would still have passed	Rejected. Job status comes from employer boards, which list every open job
Jaro-Winkler similarity	Scored different jobs of one employer as near-identical because they share a prefix ("Assistant Manager – ...")	Rejected in favour of Jaccard word overlap, chosen from labelled pairs
Lever and Recrutee	Too few Saudi postings through their APIs (14 and 17)	Not collected directly; their jobs still arrive through Jooble
Weekly movement (Q7–Q8)	Measured on employer boards, where a missing job proves it closed; one week (27 Sep) so far with all six sources fully pulled	Search-engine jobs count in every other question; trends grow as the runner is scheduled
Sparse fields	Sources often omit experience level (known for 39% of jobs) and employment type (35%)	Each answer shows its coverage; inferred levels are labelled

6 · NEXT STEPS

What we would improve next

1. **Schedule the runner weekly** (for example with Azure Data Factory or a cron job), so Q7 and Q8 become trends instead of two data points.
2. **Classify Arabic job titles** by adding Arabic keywords to the category and seniority seeds, shrinking the "Other" group.
3. **Label more matching pairs** to raise matching recall while keeping precision high.
4. **Add employer boards** for large Saudi employers, which widens the jobs with a tracked open/closed status.
5. **Track skills over time** and publish a skills-trend page for career centres.